

CITED

The Operating System for Digital Authority in AI Search

An enterprise framework for SEO, AEO, GEO, and SXO

How organizations get read, trusted, and cited when machines answer the questions

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Executive summary

Search stopped being a list of links. On a growing share of queries, an engine reads the web, writes one answer, and names a few sources. The prize is no longer a ranking that earns a click. It is a citation inside the answer. **The two are won differently, and most teams are still optimizing for the first.**

The shift is measurable. The share of United States Google searches that send no visit to the open web reached about 68 percent in early 2026. Around 73 percent of business buyers now use AI tools during research. And the engines disagree on whom to trust: ChatGPT and Perplexity overlap on only about 11 percent of the domains they cite for the same prompts. A brand absent from the answer is absent at the moment the decision gets made.

CITED is an operating model for that world. It has five pillars, run in order: **Citation Surface, Information Architecture, Technical Foundation, Evidence-Rich Extraction, and Demand Capture.** Each pillar addresses one stage of how an engine finds, reads, trusts, and cites a source, and each reinforces the next. Run together, they form a compounding loop the paper calls the Citation Flywheel.

This document makes the case for CITED as a system rather than a set of tactics, defines the mechanism behind citation, works through each pillar with its business case and evidence, sets out a five-level maturity model, and gives an operating model an enterprise team can run: a roadmap, a cadence, a governance structure, a measurement scorecard, and a map onto the SEO work already in flight.

Sources for every figure in this paper are listed, graded by tier, in the evidence section at the end.

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1 Why CITED exists

Search changed shape

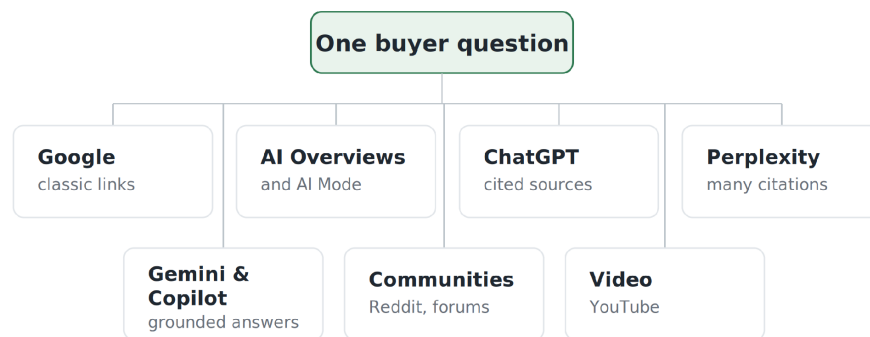
For twenty years the model held: rank on the results page, earn a click, run the visitor through a funnel, convert. Every SEO playbook optimized links in that chain. The chain now breaks at the first link. Answer engines read several sources, synthesize one response, and attach a short list of citations. Google AI Overviews, ChatGPT, Perplexity, and Gemini all work this way.

The numbers are not subtle. Zero-click searches in the United States reached about 68 percent in early 2026, the fastest move in a decade, and searches that surface an AI Overview are zero-click roughly 83 percent of the time. Visibility and traffic have come apart. A page can rank and still lose the click to an answer written above it.

SparkToro and Datos, 2024 and 2026 zero-click studies. Similarweb AI Overview click data via Leapd, 2026.

The Search Fragmentation Map

One buyer question now gets answered across many surfaces.



If you are absent from a surface, you are absent from the decisions made there.

One buyer question is now answered across many surfaces, each selecting sources its own way.

Where traditional SEO stops working

Traditional SEO assumes a page competes as a whole, on one engine, for a ranked position a human clicks. Three of those assumptions no longer hold.

- **Pages lose to passages.** Engines retrieve and cite self-contained passages, not whole documents. A strong section can carry a weak page in, and a strong page with no extractable answer stays out.
- **Owned loses to earned.** Engines corroborate across sources. Research from Toronto finds a strong bias toward earned, third-party content over brand-owned pages, with the widely cited gap near 325 percent more citations for earned media.
- **One engine loses to many.** The engines do not agree. A single flat AI strategy underperforms on all of them, because each weights sources differently.

Chen and Koudas, 2025, arXiv 2509.08919. Averi and Superlines cross-platform overlap, 2026.

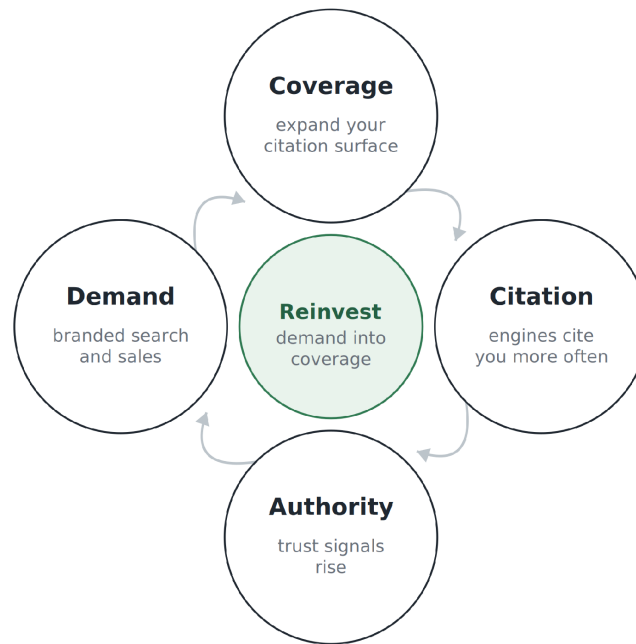
CITED is an operating system, not a campaign

A campaign ends. An operating system runs. CITED is built as the second because citation is not a one-time win. Every citation adds corroboration, which raises the odds of the next citation, which builds

the entity authority that makes new content cite faster. The work compounds, so the framework is organized to be run continuously, measured, and improved, not executed once and filed.

The Citation Flywheel

A loop where each stage raises the next, and each pass costs less.



The Citation Flywheel. Each pillar raises the next, and each turn of the loop costs less than the last.

2 How a generative engine decides what to cite

To be cited, it helps to know how the citation gets made. An engine does not rank a page and stop. It runs a sequence, and each step filters the field. Three forces decide the outcome.

Retrieval decides who is eligible

The engine rewrites the query into sub-queries and pulls a candidate set of passages from an index or a live fetch. Classic signals still gate this step. One analysis of 432,000 keywords found that 97 percent of AI Overviews cite at least one source from the top 20 organic results, so ranking remains the entry ticket. Retrieval works on passages, so structure decides what qualifies.

Selection decides who is chosen

From the candidate set, the model chooses what to quote. It favors content that is specific, consistent in style, and easy to attribute: direct answers stated up front, named sources, numbers, and clean definitions. Controlled tests show that adding statistics, quotations, or cited sources raises the chance of being used by 30 to 41 percent, while keyword stuffing scores about 9 percent below an unedited baseline.

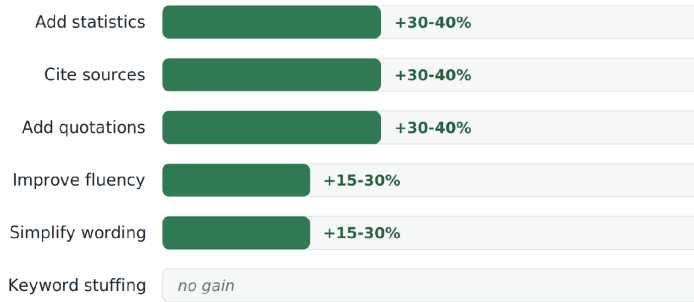
seoClarity via Leapd, 2026. Aggarwal et al., GEO, ACM SIGKDD 2024, arXiv 2311.09735.

Attribution decides who gets credit

The engine writes the answer and names a few sources. Whether it links you, recommends you, or only mentions you depends on how corroborated and trusted your entity is across the surfaces it read. This is where earned presence and entity consistency pay off.

What Actually Moves AI Citation

Relative visibility lift in generative answers, by tactic. Higher is better.



Source: Aggarwal et al., GEO, ACM SIGKDD 2024. Ranges are relative improvement.

What moves selection. Relative visibility lift by content tactic, from the GEO experiments.

Two implications follow, and they shape the whole framework. First, retrieval and selection reward different things, so you have to win both. Second, because engines lift passages and corroborate across sources, authority is built at the level of the passage and the entity, not the page. CITED is organized around exactly those levers.

3 The CITED framework, pillar by pillar

CITED has five pillars, read top to bottom, C to D. The order is a dependency. A page no crawler can read is worth nothing, so technical access sits under content. Content nobody trusts gets ignored, so authority work comes early. You can improve several pillars at once, but when you start, you fix what blocks the engines first.

The CITED Framework

Five layers, in reading order. Start at the top, with C.



The five layers, in reading order.

C Citation Surface

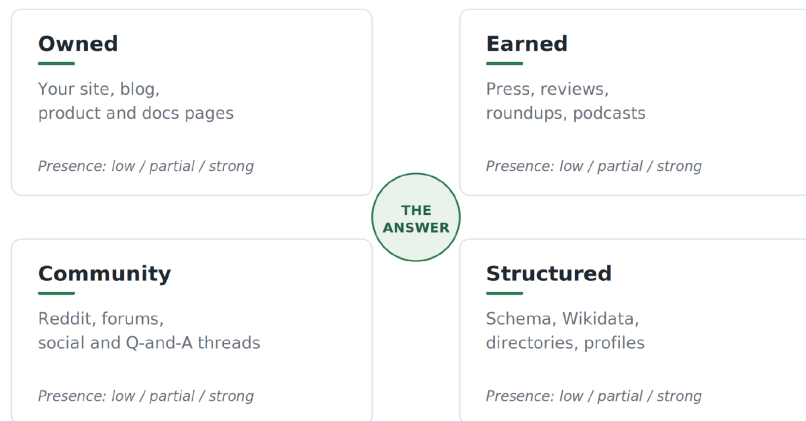
Definition. Your total footprint across the sources an engine reads: your own pages, earned media, community discussion, and structured references. A single great page is a small surface. A brand corroborated across many trusted sources has a large one.

Why it matters. Engines decide what to trust by corroboration. The more independent, credible sources say the same thing about you, the more likely you are named in the answer. This is the pillar most brands underweight, because it lives largely off their own domain where they feel least in control.

The enterprise move. Map where your category is discussed, correct the sources that describe you wrong, keep your entity consistent across Wikidata, directories, and profiles, and earn mentions in the third-party sources the engines actually pull from.

The Citation Surface Audit

Four zones an answer about you is built from. Score your presence in each.



The four zones an answer about you is built from. Score your presence in each.

EVIDENCE

Earned, third-party sources draw roughly 325 percent more citations than brand-owned pages, and brand mentions track AI citation about three times more strongly than backlinks.

Chen and Koudas, 2025. Ahrefs via Position Digital, 2026.

I Information Architecture

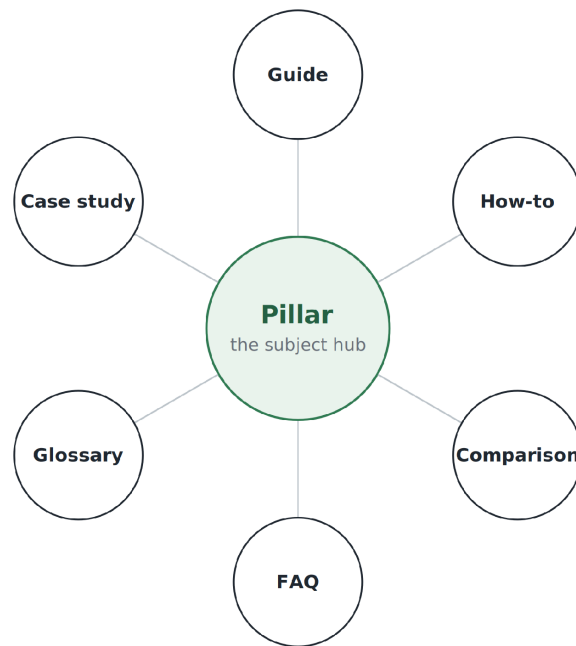
Definition. How cleanly your knowledge is organized into retrievable, self-contained units: clear headings, logical topic clusters, internal links that signal relationships, and passages that still make sense when lifted out of the page.

Why it matters. Engines retrieve passages, not pages, and they read structure as a signal of depth. A well-architected cluster tells the model you own a subject, which raises the odds it treats you as the default source for it. Narrow, per-query optimization hits a ceiling near 70 percent effectiveness and overfits one question while losing the neighbors.

The enterprise move. Choose bounded subjects to own. Build hub-and-spoke clusters with one pillar linking to every subtopic and back. Remove pages that compete with each other. Make each passage answer one question cleanly.

The Hub-and-Spoke Blueprint

One pillar, many supporting pages, all linked both ways.



One pillar, many supporting pages, linked both ways. Depth an engine can navigate.

EVIDENCE

Structural and semantic revision raises a source's measured influence on answers more than word-count tactics do.

Chen X. et al., 2025. Chen Q. et al., CC-GSEO-Bench, 2025.

T Technical Foundation

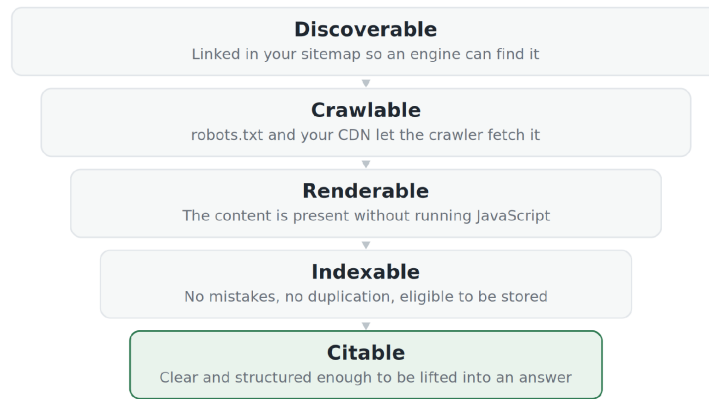
Definition. The plumbing that lets the other pillars work: crawler access, clean server-rendered HTML, fast load, structured data, and correct indexation. If a bot cannot reach and read your content, nothing above this matters.

Why it matters. AI crawlers split by job. Training crawlers such as GPTBot and Google-Extended collect data for future models. Retrieval crawlers such as OAI-SearchBot and PerplexityBot fetch pages to answer a live question. Block the retrieval crawlers and you drop out of the answers, whatever your ranking. Most AI crawlers also do not run JavaScript, so content painted in by script is invisible to them.

The enterprise move. Allow the retrieval crawlers you want. Serve content in HTML, not script. Pass Core Web Vitals for real users. Mark up content with schema. Treat llms.txt as optional and unproven; Google said in 2025 it does not use it, and controlled tests show no citation lift.

The Crawl-to-Index Funnel

A page must clear every stage to be cited. Most sites leak at one of them.



Fix the highest broken stage first. Everything below it depends on it.

A page must clear every stage to be cited. Most sites leak at one of them.

EVIDENCE

Freshness is a real signal: 85 percent of the sources cited in AI Overviews were published in the last two years.

Ahrefs via Position Digital, 2026. No Hacks AI user-agent reference, 2026.

E Evidence-Rich Extraction

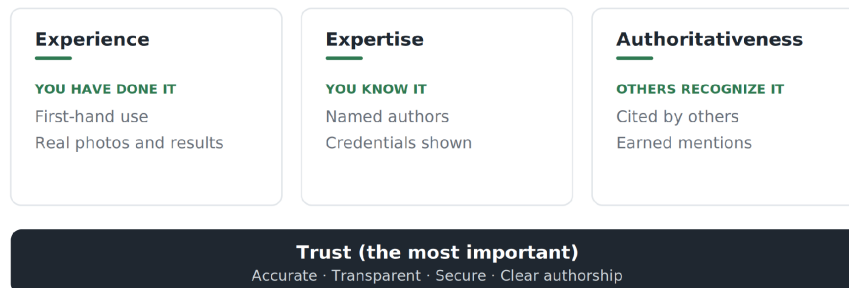
Definition. Writing so that every key claim is quotable, attributable, and hard to argue with: the direct answer first, then specific numbers, named sources, and proof that a real expert stands behind the page.

Why it matters. Retrieval gets you into the pool. Extraction gets you chosen. Engines will not safely lift vague, hedged, generic text, so it rarely gets cited. As AI floods the web with confident, unsourced prose, the scarce and valuable thing is verifiable proof of real experience.

The enterprise move. Lead each page with the answer. Replace vague claims with numbers. Cite credible sources. Show named authors and their credentials. Add first-hand evidence: original data, real results, real photos. Treat every page as a small case you have to prove.

The E-E-A-T Evidence Ledger

Prove a real human stands behind the page. Trust holds the rest up.



Prove a real human stands behind the page. Trust holds the rest up.

EVIDENCE

Adding statistics, quotations, or cited sources raised generative-answer visibility by 30 to 41 percent. Cite Sources alone lifted a fifth-ranked page by 115 percent, and pages with expert quotes average 4.1 citations against 2.4 without.

Aggarwal et al., KDD 2024. SE Ranking via Foglift, 2026.

D Demand Capture

Definition. The bridge from citation to revenue. It makes sure the pages engines cite move a reader toward action, that the branded search lift from AI mentions is captured, and that visibility is measured all the way to pipeline.

Why it matters. Visibility without conversion is vanity. A citation that sends a curious buyer to a dead-end page is a wasted win. The economics reward the brands that close this loop.

The enterprise move. Lead cited pages with one clear next step. Capture branded demand created by AI mentions. Expose machine-readable pricing and availability for agents. Track from presence to citation to demand to revenue, and report the last number to finance.

The Visibility-to-Value Dashboard

Four questions, from showing up to making money. Measure all four.



Most teams stop at presence. The money is in tracking all the way to revenue.

Four questions, from showing up to making money. Measure all four.

EVIDENCE

AI-referred visitors converted at about 14.2 percent against 2.8 percent for Google organic, and the top citation slot carries roughly a 23 times conversion premium, with cited brands earning about 35 percent more organic and 91 percent more paid clicks.

The Digital Bloom via Foglift, 2026.

4 The CITED maturity model

A framework is only useful if you can place yourself on it. This model works like a Gartner or CMMI ladder. Find your level, then work the pillar gap that moves you up. Most organizations sit at level 2 or 3 and do not realize how much citation they are leaving on the table.

Level	Profile	What it looks like	The move up
1 Absent	No presence	Engines describe you from stale third-party data. You appear in few answers, some wrong. Crawlers may be blocked or content does not render.	Fix Technical Foundation. You cannot be cited if you cannot be read.

Level	Profile	What it looks like	The move up
2 Reactive	Fixes after the fact	You correct profiles and add schema once a problem shows up. Citations are occasional and accidental. Competitors are cited more.	Invest in Information Architecture. Organize content into clusters and retrievable passages.
3 Present	Runs the system	You run the five pillars and appear for your core topics, but as one option among many, rarely the primary source.	Strengthen Evidence-Rich Extraction. Make your passages the most specific, credible answer.
4 Cited	A trusted source	Engines name you across several platforms. Citations start to lift branded search and pipeline.	Close the loop with Demand Capture. Turn visibility into measured revenue and widen your surface.
5 Compounding	Self-reinforcing	You are a default cited authority. New content earns citations fast because your entity is already trusted, and AI visibility is an attributed source of pipeline.	Defend and extend. Document authority so it stays citable, and hold the lead.

The dimensions that move with each level are the same five the framework names: how broad and corroborated your surface is, how well your content is structured, how reachable and fast your pages are, how specific and sourced your claims are, and how tightly visibility connects to revenue. The self-scorecard in the measurement section turns this ladder into a number out of 25.

5 The operating model

CITED runs as an operating discipline. Below is the structure an enterprise team can stand up: a staged rollout, a fixed cadence, clear ownership, and a governance loop.

Rollout

Window	Focus	Outcome
Days 1 to 30	See and map. Audit your citation surface across buyer questions. Find where you are cited now and by whom.	A baseline and a ranked gap list.
Days 31 to 90	Build. Open crawler access and rendering, write CITE-ready content, claim and correct mentions, ship your first clusters.	Reachable, structured, evidence-rich pages.
Days 91 to 180	Capture and measure. Add per-engine moves and capture mechanisms. Stand up the dashboard and report to finance.	Attributed visibility, first revenue signal.
Days 181 to 365	Compound. Widen surface, refresh on schedule, defend the lead, and run the flywheel.	A self-reinforcing citation position.

Cadence and governance

Rhythm	What happens	Owner
Weekly	Publish and refresh on schedule. Reply to reviews. Watch citation and referral dashboards for movement.	Content and SEO leads
Monthly	Re-run buyer prompts across engines. Update the citation gap list. Report presence, citation, and demand.	AI visibility owner

Rhythm	What happens	Owner
Quarterly	Re-score the five pillars. Audit crawler access and rendering. Reset priorities to the lowest pillar.	Head of growth
Continuous	Prune weak pages, correct wrong sources, and feed measured wins back into the surface.	Whole team

Ownership

CITED crosses functions, so it needs a named owner and a light RACI. One person owns AI visibility as a number. Content owns extraction and architecture. Engineering owns technical access and rendering. PR and community own earned surface. Analytics owns the measurement chain. Leadership owns the priority call each quarter. Without a single accountable owner, the work scatters and the flywheel never starts.

6 CITED across AEO, GEO, and SXO

CITED is one system that reads across the labels the industry uses for this shift. Each label is a lens on the same mechanism, and CITED maps cleanly onto all three.

CITED and answer engine optimization

AEO is about being the answer engines extract and present: featured snippets, AI Overviews, People Also Ask, voice, and multi-turn follow-ups. CITED serves it directly. Information Architecture makes content chunkable into clean passages. Evidence-Rich Extraction gives the engine a direct, sourced answer it can lift. Technical Foundation makes the passage reachable. The unit of work is the self-contained answer, not the page.

CITED and generative engine optimization

GEO is about visibility inside generated answers across ChatGPT, Gemini, Perplexity, Copilot, and AI Mode. The engines trust different sources, so you optimize per engine, not for AI as one thing. Citation Surface widens the corroboration each engine reads. Evidence-Rich Extraction raises selection odds. Demand Capture turns the citation into pipeline.

The Four-Engine Playbook

Each engine trusts different sources. Optimize per engine, not for AI as one thing.

ChatGPT

Leans on encyclopedic and review sources

BACKED BY
Its own index plus Bing

Perplexity

Leans on Reddit and real-time community

BACKED BY
Crawls the web continuously

Google AI Overviews

Leans on YouTube and multimodal content

BACKED BY
Gemini and Google's index

Microsoft Copilot

Leans on the Bing index and schema

BACKED BY
Structured data helps it read you

Only about 11% domain overlap between ChatGPT and Perplexity

Each engine trusts different sources. Optimize per engine, not for a single AI.

CITED and search experience optimization

SXO is about what happens after the click: trust, information scent, task completion, decision confidence, and the behavioral signals that follow. CITED feeds it. Evidence-Rich Extraction builds credibility on the page. Demand Capture leads the reader to a clear next step. A cited page that fails the human who arrives wastes the citation, so experience is part of the system, not a separate concern.

7 Overlay on existing SEO

CITED does not replace SEO. It overlays it, and points the same work at citation as well as ranking. The mapping below shows how core SEO areas change under the framework.

SEO area	How CITED enhances it	Primary KPI
Technical SEO	Adds AI crawler access and server-side rendering as first-class requirements.	Retrieval eligibility rate
Content strategy	Shifts the unit of work from the page to the extractable passage and the owned subject.	Citations per topic
Keyword research	Becomes buyer-question mapping across engines, not just search-volume targeting.	Prompt coverage
Topical authority	Formalized as hub-and-spoke clusters engines read as depth.	Share of citation on a topic
Internal linking	Signals cluster relationships that help retrieval find the right passage.	Cluster coverage
Schema	Classifies content so machines parse and trust it.	Rich-result and entity coverage
EEAT	Operationalized as Evidence-Rich Extraction with named authors and proof.	Cited-source rate
Digital PR	Reframed as earned Citation Surface, the strongest corroboration signal.	Earned mentions in cited sources
Entity SEO	Consistent Wikidata, profiles, and knowledge-graph presence engines can resolve.	Entity consistency score
Content refresh	A scheduled habit, since 85 percent of AI Overview citations are recent.	Freshness ratio
Local SEO	Complete profiles, review velocity, and consistent citations for pack and AI eligibility.	Local pack and AI presence
Measurement	Extended from rankings and clicks to presence, citation, demand, and revenue.	AI-influenced pipeline

8 Measurement: the CITED scorecard

Most teams measure presence and stop. The money is in tracking all the way to revenue. CITED measures four tiers, and the gap between them is where the story lives.

Tier	The question	What to track
Presence	Are you in the answer at all?	Appearance rate across buyer prompts, per engine
Citation	Are you cited, with a link?	Citation rate, share of citation on a topic

Tier	The question	What to track
Demand	Are people acting?	AI referral traffic and conversion, branded search lift
Revenue	Is it money?	AI-influenced pipeline and revenue, conversion premium

Underneath the tiers, score the five pillars from 1 to 5 on the self-scorecard: Citation Surface, Information Architecture, Technical Foundation, Evidence-Rich Extraction, and Demand Capture. Add them for a total out of 25. Below 11, fix access and structure. From 11 to 16, work extraction. From 17 to 21, tighten demand capture and widen surface. Above 21, defend the lead. Your lowest pillar is always the highest-leverage next move. Re-score quarterly, because the ground moves.

Leading indicators tell you the work is landing before revenue shows: retrieval eligibility, citation rate, earned mentions, freshness. **Lagging indicators** confirm it paid: AI referral conversion, branded search, AI-influenced pipeline. The north-star is share of citation on the questions your buyers actually ask.

9 How CITED compares

CITED does not discard the models that came before it. It absorbs them and adds the citation layer they predate.

Model	What it gives	What it misses	How CITED unifies it
Traditional SEO	Ranking and crawlability	Assumes rank equals traffic	Keeps access, aims it at citation
EEAT	Trust as a quality bar	Guidance, not a build system	Operationalized as the E pillar
Topic clusters	Topical depth	Page-level, not passage-level	Becomes the I pillar, tuned for retrieval
Hub-and-spoke	Internal structure	Silent on off-domain trust	Structure plus Citation Surface
Entity SEO	Machine-readable identity	Narrow slice of the problem	Folded into I and C
Content marketing	Volume and reach	Not built for extraction	Rewritten for evidence and citation
Digital PR	Earned coverage	Treated as awareness, not signal	Reframed as Citation Surface
Brand SEO	Branded demand	Downstream only	Closed as Demand Capture

The pattern is consistent. Each prior model owns one part of the answer. CITED sequences them into one operating loop and adds the part they all predate: earning the citation itself.

10 The CITED framework library

The system carries a set of working models. Each has a job, an input, and an output.

- **The CITED operating system.** The five pillars run in dependency order. Input: your current state. Output: a prioritized plan.

- **The Citation Flywheel.** The compounding loop from surface to citation to authority to demand and back.
- **The Citation Surface Audit.** Four zones, owned, earned, community, and structured, scored for presence.
- **The Search Fragmentation Map.** One buyer question across many surfaces, used to plan per-engine coverage.
- **The Crawl-to-Index Funnel.** Five access stages a page must clear to be citable, used to find the leak.
- **The EEAT Evidence Ledger.** The proof each trust signal needs on the page.
- **The Four-Engine Playbook.** Per-engine source preferences for ChatGPT, Perplexity, Google, and Copilot.
- **The Visibility-to-Value Dashboard.** Presence, citation, demand, and revenue in one measurement stack.
- **The CITED Maturity Model and Self-Scorecard.** A five-level ladder and a 1-to-5 score per pillar to place and track a team.

The case, in one line

Ranking was the game when a ranking earned a click. That link is broken for a growing share of search, and the new game is citation, which is won by being read, trusted, and corroborated at the level of the passage and the entity. CITED is the operating system for that game. Run it as a discipline, not a campaign, and the advantage compounds in a way competitors struggle to catch.

Evidence

Figures are graded in three tiers: peer-reviewed and preprint research, then primary measurement, then vendor compilations, which are directional. Trace each to its source before quoting it in a formal setting.

Tier one: peer-reviewed and preprint

Aggarwal, P. et al. (2024). GEO: Generative Engine Optimization. ACM SIGKDD 2024. arxiv.org/abs/2311.09735

Chen, M., Koudas, N. et al. (2025). Generative engine optimization: how to dominate AI search. arxiv.org/abs/2509.08919

Chen, X. et al. (2025). Role-augmented intent-driven generative search engine optimization. arxiv.org/abs/2508.11158

Chen, Q. et al. (2025). Beyond keywords: content-centric agents for generative engine optimization. arxiv.org/abs/2509.05607

Rosu, M. (2025). Authority versus topical authority in the world of AI. doi.org/10.47743/eufire-2025-1-27

Tier two: primary measurement

SparkToro and Datos (2024, 2026). Zero-click search studies. sparktoro.com

Profound (2025). AI platform citation patterns. tryprofound.com/blog/ai-platform-citation-patterns

Tier three: vendor compilations, directional

Averi and Superlines (2026). Cross-platform citation overlap. authoritytech.io

Leapd (2026). How ChatGPT, Google AI Overviews, and Perplexity source information. leapd.ai

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Position Digital (2026). AI SEO statistics, compiling Ahrefs and Profound. position.digital/blog/ai-seo-statistics

The Digital Bloom (2026). AI citation position and revenue report. via foglift.io

No Hacks (2026). The AI user-agent landscape. nohacks.co/blog/ai-user-agents-landscape-2026

About the author

Avik Bal is a B2B digital growth strategist working across search, content, and AI visibility for software and enterprise brands. CITED is his framework for earning citation in AI search. He is based in Bengaluru, India, and works with teams and founders worldwide.